GENETICS LEARNING FRAMEWORK

The **Genetics Learning Framework** includes a broad overview of topics within the field of genetics and describes a comprehensive genetics curriculum for an undergraduate biological sciences student. The framework provides a starting point for course and curriculum design, identifying concepts that could be included. A single genetics/biology course would typically not cover all these topics, but choose focus areas within this framework that are most relevant to their context and student population.

This revised framework is designed for relevance to genetics students broadly, including majors and non-majors. Although targeted primarily at an undergraduate student population, it can be adapted for high school or graduate student levels. Updates include increased emphasis on biological diversity and the range of genetic systems employed by different organisms, as well as a new focus on the social and societal impacts of the way we describe and teach genetics principles.

The framework is organized in three levels: foundational concepts, core concepts, and sample learning outcomes:

- **Foundational Concepts** identify large areas within the field of genetics. These concepts help to organize different types of information related to genetics. Some of them contain more subtopics than others and may require correspondingly more instructional time.
- **Core Concepts** provide a more detailed and comprehensive overview of the topics within each Foundational Concept. The descriptions within each Core Concept encompass a complete undergraduate genetics curriculum, though individual courses would typically not include every component. Depending on the course context, certain aspects of the Core Concepts may be selected as a focus. Descriptions of the Core Concepts reflect long-term areas of inquiry in genetics to provide a stable framework, and do not identify methodologies or newly emerging concepts that may quickly become outdated.
- **Sample Learning Objectives** include representative examples that could be used in a course, but they are not designed to cover the entire scope of learning objectives in that area. The samples provided include a range of difficulty levels, Bloom's levels, and types of biological thinking across the Core Concept area. These learning objectives are intended to provide models for the types of learning students might be asked to do in various contexts, with the expectation that instructors will need to add more learning objectives for their specific course, and may not include all the listed examples in any single course. Written as student-facing resources, learning objectives should be preceded by "Students should be able to…"

A complete genetics curriculum should include building skills in the process of science related to genetics, as well as exploration of genetics concepts. A Learning Framework for Science Process Skills is available separately through CourseSource and can be applied to the genetics concepts in this framework.

Overview

Nature of Genetic Material

- **How is DNA organized?**
- **What are the molecular components and mechanisms necessary to preserve and duplicate an organism's genome?**

Transmission and Patterns of Inheritance

- **What are the mechanisms by which an organism's genome is passed on to the next generation?**
- **How do interactions among genes, alleles, and the environment result in a variety of inheritance patterns and phenotypic outcomes?**
- **How does the phenomenon of linkage affect the assortment of alleles during meiosis and the associations of common variants in a population?**

Molecular Biology of Gene Function

- **How is genetic information expressed and how does it affect an organism's structure and function?**
- **How do different types of variations affect genes and the corresponding mRNAs and proteins?**

Gene Expression and Regulation

- **How does the control of gene expression lead to changes in an organism's structure and function?**
- **What mechanisms are used by cells to regulate the expression of genes?**
- **How does systems biology contribute to understanding the genetic basis of complex traits where many genes interact with environmental factors?**

Evolution and Population Genetics

- **What are the processes that can affect the frequency of genotypes and phenotypes in a population over time?**

Methods and Tools in Genetics

- **How do studies in model organisms help us understand fundamental aspects of genetics?**
- **What experimental approaches are commonly used to analyze gene structure, gene expression, gene function, and genetic variants?**

Genetics and Ethics

- **How does genetics impact society and society impact genetics?**

Foundational Concept: Nature of Genetic Material

Core concept: How is DNA organized?

DNA, the hereditary molecule of life, is composed of nucleotides, which form complementary, antiparallel strands in a double helix. DNA strands can vary in length, be linear or circular, and vary in copy number. DNA can be further compacted into chromosomes by proteins, including histones. Specific segments of DNA function as genes that encode RNAs or proteins, and other sequences have regulatory functions or other important roles. The entire genetic material of an organism is known as its genome, which is organized into different numbers of chromosomes depending on species/cell types. Some organisms (or cell types within a multicellular organism) have multiple copies of each chromosome and the number of homologous chromosomes present determines the cell's ploidy. Some organisms also contain DNA in extrachromosomal forms, such as plasmids. Viruses contain genomes that take a variety of forms and may be composed of DNA or RNA.

Sample Learning Objectives:

- Compare the structure of DNA and RNA, and discuss how the structure of each molecule contributes to its function.
- Analyze how the size, shape, and charge of nucleotides contribute to their ability to pair.
- Summarize the experimental evidence that supported the identification of nucleic acids as the hereditary molecule.
- Explain how the structure and composition of DNA allow it to perform its function as the genetic material of the cell.
- Describe how DNA is packaged in eukaryotic chromosomes in terms of histones, nucleosomes, and chromatin.
- Describe how DNA is packaged in prokaryotic organisms, including extrachromosomal forms.
- Explain and provide examples of ways in which variations in ploidy can have implications for cell specialization and for reproduction/speciation.
- Explain the meaning of ploidy (haploid, diploid, aneuploid, etc.) and how it relates to the number of homologues of each chromosome.
- Describe the genomic structure (genome size, chromosome number, etc) of various organisms and explain the lack of correlation between chromosome size and genetic information content.
- Predict the potential impact of specific extrachromosomal DNA elements (e.g., antibiotic resistance plasmids) on microbial populations or eukaryotic cell processes.
- Describe the types of nucleic acid that viruses can have. Compare and contrast the functional significance of DNA vs RNA as viral genetic material.

Core concept: What are the molecular components and mechanisms necessary to preserve and duplicate an organism's genome?

DNA is replicated by enzymes that synthesize new DNA strands, using the original strands as templates, in a semi-conservative manner. Typically, DNA replication precedes cell division, and copies of the genome are

distributed to progeny cells. Proofreading and repair mechanisms safeguard the accuracy and integrity of the replicated genome, however when changes to the DNA sequence are introduced this alters the information stored in that cell and its progeny.

Sample Learning Objectives:

- Summarize the experimental evidence that supported the semiconservative nature of DNA replication, and be able to follow a labeled DNA molecule through several rounds of replication.
- Describe the steps of DNA replication in prokaryotes, highlighting enzymes that are active at each step.
- Contrast DNA replication in eukaryotic cells with that in prokaryotic cells, describing additional steps that are necessary in eukaryotic cells.
- Predict the consequences when the functions of DNA replication enzymes are altered.
- Draw, analyze, and explain the structure of DNA replication bubbles and replication forks, including leading strand/lagging strand, Okazaki fragments.
- Explain how the directionality of DNA polymerase and the antiparallel structure of double stranded DNA are responsible for the differences between the leading and lagging strands.
- Discuss the evolutionary significance of germline vs. somatic mutations.
- Differentiate between major DNA repair pathways (e.g., nucleotide excision repair, base excision repair, mismatch repair, nonhomologous end joining), how they resolve DNA damage, and explain their importance in maintaining genetic integrity.

Foundational Concept: Transmission and Patterns of Inheritance

Core concept: What are the mechanisms by which an organism's genome is passed on to the next generation?

An organism's genome is passed on to the next generation through the processes of binary fission, mitosis, and/or meiosis. In each of these cases, the genome is first replicated and then distributed to progeny cells. Binary fission is the predominant process used by prokaryotes for asexual reproduction and results in two nearly identical progeny cells. Mitosis also generates two nearly identical cells and is used in asexual reproduction in eukaryotes, in addition to growth, development, and tissue repair in multicellular organisms. Meiosis is used in sexual reproduction in eukaryotes and includes two rounds of cell division, resulting in cells with reduced ploidy relative to the parent. Meiosis also increases genetic variation due to independent assortment of chromosomes and through crossing over, a process that exchanges genetic material between homologous chromosomes. Either directly after meiosis or later in the lifecycle, the reduced-ploidy cells generated by meiosis develop into gametes and fuse to form a zygote with a new combination of genetic material. In addition to genetic information that is passed from one generation to the next, genetic information can also be passed horizontally and between organisms, for example plasmid exchange in prokaryotes, genetic material introduced by viruses, and transposable genetic elements.

Sample Learning Objectives:

- Distinguish between somatic and germline cells; listing similarities and differences.
- Describe, using diagrams, the sequence of events involving DNA in meiosis from chromosome duplication through chromosome segregation.
- Explain how meiosis is different from mitosis.
- Distinguish between sister chromatids and homologous chromosomes; describe and draw how each is involved in mitosis and meiosis.
- Differentiate between a gene and an allele.
- Describe Mendel's laws of Segregation and Independent Assortment, and their origins in meiosis.
- Explain how independent assortment of alleles during meiosis can lead to new combinations of alleles of unlinked genes.
- Calculate the probability of a particular gamete being produced from an individual, assuming independent segregation.
- Calculate the probability of a particular genotype, given independent segregation and random union of gametes between two individuals.
- Explain how crossing over (homologous recombination) can lead to a new combination of alleles on a chromosome.
- Explain how checkpoints help to regulate the cell cycle, including identifying points in the cell cycle that are guarded by checkpoints and describing the negative consequences to cells that continue through the cell cycle inappropriately.
- Discuss how unequal chromosome numbers can arise during meiosis, and why such alterations can be detrimental, neutral, or beneficial.

Core concept: How do interactions among genes, alleles, and the environment result in a variety of inheritance patterns and phenotypic outcomes?

Variants in a gene sequence are referred to as alleles, and the combination of alleles (genotype) present in a cell/organism contributes to its characteristics (phenotypes). Most characteristics of an organism are due to the effects of multiple genes and contributions of environmental factors. In some cases, patterns of inheritance can demonstrate whether a trait is dominant or recessive, whether it is controlled by a single gene or multiple genes, whether it is transmitted on an autosome or a sex chromosome (allosome), and whether multiple genes are linked or assort independently. Inheritance patterns can also be influenced by additional factors including epistasis, incomplete penetrance, variable expressivity, incomplete or co-dominance, epigenetics, and non-nuclear inheritance. Understanding the genetic basis of traits, can provide insights into the underlying molecular mechanisms of gene function.

Sample Learning Objectives:

- Given the mode of inheritance for alleles of a gene, predict phenotype from genotype.
- Apply concepts of dominant and recessive inheritance in problems involving monohybrid, dihybrid, and test crosses.
- Use probability rules to predict proportions of genotypes and phenotypes of expected offspring in crosses involving one and two unlinked genes.

- Provide examples of mechanisms by which two individuals with identical genotypes might display different phenotypes.
- Contrast the mechanisms of inheritance of nuclear and organellar genetic information.
- Given a pedigree, identify the likely mode of inheritance or calculate the probability that an individual is a carrier and/or that offspring will be affected or carriers.
- Compare observed offspring of a cross to predicted ratios, using a chi-square test.
- Define penetrance, expressivity, and pleiotropy, and compare and contrast these phenomena.
- Define epistasis, and explain how interactions between the products of two genes might lead to deviations from expected Mendelian ratios in a dihybrid cross.
- Evaluate how genes and the environment can interact to produce a phenotype, and apply these principles to examples.
- Define maternal-effect lethality, and describe how maternally-inherited gene products influence phenotypes.
- Distinguish among inheritance patterns of sex-limited, sex-influenced, and sex-linked traits.
- Explain how continuous traits are the result of many different gene combinations that can each contribute a varying amount to a phenotype.
- Compare genetic data of different racial groups to demonstrate there is a roughly equal amount of genetic variation within racial groups as there is between them.

Core concept: How does the phenomenon of linkage affect the assortment of alleles during meiosis and the associations of common variants in a population?

Linked genes tend to be inherited with the same allele combinations together, as they were present in the parent organism. This phenomenon is due to the physical proximity of the genes on the chromosome, which reduces the likelihood of recombination between them. Recombination, which generates new combinations of alleles on a chromosome, results from crossing over, a process that exchanges genetic material between homologous chromosomes during meiosis and helps to coordinate meiotic chromosome segregation in many systems. Linkage can be observed in patterns of inheritance and can be used as a tool for studying genes of interest, for example in genome-wide association studies and polygenic risk scores.

Sample Learning Objectives:

- Diagram the process of homologous recombination during meiosis and explain how it can lead to new combinations of linked alleles.
- Explain the role of homologous recombination in ensuring proper segregation of homologs in meiosis I.
- Explain how a specific combination of linked alleles (haplotype) can persist through many generations (linkage disequilibrium).
- Calculate gene linkage, genetic map distances, and interference from the frequencies of progeny with recombinant phenotypes from genetic crosses.
- Explain how genetic distance is different from physical distance.
- Calculate the probability of a particular gamete being produced from an individual, provided map distance.

- Design a genetic cross or crosses that would distinguish between linked and unlinked genes.
- Evaluate the strengths and limitations of genome-wide association studies, including the significance of genetic linkage in these studies.
- Interpret LOD scores to determine the likelihood of genetic linkage between a variant of interest and a known marker, and be able to apply this information to a genetic counseling context.

Foundational Concept: Molecular Biology of Gene Function

Core concept: How is genetic information expressed and how does it affect an organism's structure and function?

The information encoded in DNA ultimately determines the genetic contribution to the structure and function of an organism by specifying both the proteins and the non-coding RNAs that are produced. To make a functional protein based on genetic information, the DNA sequence of a gene is used to generate a complementary messenger RNA (mRNA) molecule (transcription). The sequence of the mRNA coordinates tRNAs to deliver specific amino acids, which are linked together by the ribosome, in order to synthesize a particular protein (translation). Proteins carry out most functions of a cell, such as providing structural support, enzymatic activity, and cell signaling. The specific sequence of amino acids in a protein determines its three-dimensional structure and function. Not all genes or RNA molecules code for proteins. Non-coding RNAs play essential roles, including in gene expression and regulation.

Sample Learning Objectives:

- Draw a simple line diagram showing a segment of DNA from a gene and its RNA transcript, indicating which DNA strand is the template, the direction of transcription and the polarities of all DNA and RNA strands.
- Explain how the genetic code relates the information stored in the mRNA to the protein that is produced.
- Describe how expansion or retraction of triplet repeats can alter gene function and impact phenotype.
- Distinguish between the template strand and the coding strand of DNA, explain how each is related to the process of transcription and describe their similarities/differences with the RNA transcript.
- Identify changes that are made to RNA transcripts in eukaryotic cells to generate the mature mRNA, including splicing and addition of a 5' cap and a 3' PolyA tail.
- Use the genetic code to analyze a transcript sequence and identify the amino acid sequence that is encoded.
- Compare and contrast the mechanism of translation in prokaryotes and eukaryotes.
- Explain what the reading frame is, how cells use the start codon to determine which reading frame to use, and the consequences if they use the wrong reading frame.
- Describe the role of alternative splicing in generating proteins of diverse structure and function from a single gene.

- Explain the various roles of non-coding RNAs in gene expression at different levels (e.g., transcription, post-transcription, translation).
- For each step of translation, discuss major molecular events and necessary enzymes in prokaryotes, and then discuss specializations for eukaryotes.
- Describe the role of the ribosome in translation, relating structure to function.

Core concept: How do different types of variations affect genes and the corresponding mRNAs and proteins?

Variation within gene sequences, among individuals in a population, is a ubiquitous and naturally-occurring characteristic of biological systems. Differences in gene sequence can result in differences in protein structure and function. When new variants are introduced in an existing sequence they are referred to as mutations, and they can take a variety of forms. Point mutations affect single nucleotides and can be categorized as silent, missense, and nonsense mutations depending on their impact on protein structure. Frameshift mutations, including insertions and deletions, alter the reading frame of the DNA sequence. Variations in non-coding sequences can affect gene expression. The phenotypic consequences of these genetic variations differ significantly, depending on the specific gene affected, the type of variation, the organism's genetic background, environmental contributors, and other factors. The presence of variation in a population is a critical element of evolution.

Sample Learning Objectives:

- Describe how duplications, deletions, inversions, and translocations can affect gene function, gene expression, and genetic recombination.
- Describe how variation in a population arises through the process of mutation, how different types of mutations arise, and how environmental factors can affect mutation rate.
- Explain the importance of variation in populations and in evolution; cite examples of how rare or new forms can be beneficial to organisms and populations.
- Predict the most likely effects on protein structure and function of null, reduction-of-function, overexpression, dominant-negative and gain-of-function mutations.
- Compare the role of both loss-of-function and gain-of-function mutations in the origin of tumors.
- Predict, given the nucleotide sequence of a gene, the effect of a given mutation on the product of that gene, and how likely it is to affect protein function.
- Differentiate between ‘genotype’ and ‘phenotype’ and explain how both genes and the environment affect phenotype.

Foundational Concept: Gene Expression and Regulation

Core concept: How does the control of gene expression lead to changes in an organism’s structure and function?

By regulating which genes are turned on or off, and at what level they are expressed, cells can produce specific

proteins that contribute to their structure and function. For example, during multicellular development, where almost all cells in an organism's body have the same DNA, different genes are activated in different cells, leading to the formation of specialized tissues. Similarly, in response to environmental cues, cells or organisms can alter their gene expression patterns to adapt to changing conditions. This intricate regulation of gene expression allows organisms to grow, develop, and/or maintain their homeostasis.

Sample Learning Objectives:

- Define "gene regulation" at a molecular level and explain why changes in gene expression are critical for growth, development, and maintenance.
- Explain how most cells can have the same genetic content and yet have different functions in the body.
- Discuss environmental impacts on epigenetic systems.
- Describe how differential histone modification modulates gene activity and is utilized in developmental progression
- Describe genetic cascades; use the sex-determination cascade to explain how differential gene expression can result in the development of diverse sexual anatomy and reproductive organs.
- Explain how polarity is established in a developing embryo using gene expression gradients.
- Distinguish between examples of variation due to differences in genetic information encoded and due to differences in regulation of gene expression.
- Explain the process of programmed DNA elimination as a form of genetic regulation and evaluate its impact on gene expression in the development of somatic cells versus germ cells.
- Design an experiment using reporter genes or transcriptomic analysis to investigate how gene expression changes affect cellular function in a specific biological context.

Core concept: What mechanisms are used by cells to regulate the expression of genes?

Cells regulate gene expression through a multi-layered process. Chromatin remodeling affects the accessibility of DNA to transcription machinery, transcription factors bind to DNA to control the initiation of transcription, RNA processing and other factors can regulate mRNA stability and translation. Finally, post-translational modifications like phosphorylation and protein degradation can modulate protein activity and abundance.

Sample Learning Objectives:

- Discuss the potential roles of DNA modification, histone modification, and non-coding RNA in epigenetic inheritance, both somatic and germline.
- Discuss how transcription factor proteins work to activate and repress gene expression.
- Explore how epigenetic changes can be inherited across generations and their implications for the development of traits.
- Explain the roles of inducers and repressors in the expression of multiple clustered genes that are regulated by the same signal, using the example of the lac operon in bacteria.
- Describe the role of regulatory elements (e.g. sequences in DNA that are binding sites for transcription factors) in regulating gene expression. Compare and contrast regulatory elements in prokaryotes and eukaryotes.

- ❖ Compare and contrast the roles of small RNAs, microRNAs(miRs), short interfering RNAs (siRNAs) in regulating gene expression in eukaryotes.
- ❖ Describe how positive and negative regulation can contribute to differential gene or protein expression in eukaryotes, and give examples of each.
- ❖ Provide examples of how the environment affects gene expression.

Core concept: How does systems biology contribute to understanding the genetic basis of complex traits where many genes interact with environmental factors?

Systems biology is an approach that studies biological entities—like genes, proteins, and cells—not in isolation, but as an integrated network that collectively gives rise to a particular biological function or trait. This interdisciplinary field uses computational and mathematical models to understand how these parts interact. For complex traits, where no single gene is responsible, systems biology maps Gene Regulatory Networks (GRNs) and uses pathway analysis to track how changes in gene expression propagate through the system. Multifactorial analysis integrates genetics with environmental factors to identify the probability of developing certain traits. Data from populations, like those gathered through GWAS (Genome-Wide Association Studies), are integrated into these mathematical models to holistically understand these complex traits.

Sample Learning Objectives:

- ❖ Demonstrate how the interdisciplinary approach of systems biology can be used to identify potential drug targets based on complex disease mechanisms.
- ❖ Explain how genomics, transcriptomics, proteomics, and metabolomics each contribute to studies of complex biological systems.
- ❖ Analyze how genes interact within networks and pathways to regulate cellular functions. Given a specific pathway or network, predict the effects of loss-of-function or gain-of-function mutations in a specific gene.
- ❖ Explain the concept of environmental contributions to phenotypic variation and identify key categories of non-genetic factors—such as nutritional, chemical, physical, social, and developmental influences—that interact with genetic makeup to shape observable traits.
- ❖ Evaluate how gene regulatory networks, epigenetic modifications, and environmental stressors contribute to an example with societal impact, such as drought tolerance in maize, with implications for crop engineering.
- ❖ Describe the experimental evidence that supports skin color as a polyphyletic trait and is not monophyletic (for example, recent studies showing that the genetic basis for melanin production among populations in Central and South America is distinct from western European and African populations).

Foundational Concept: Evolution and Population Genetics

Core concept: What are the processes that can affect the frequency of genotypes and phenotypes in a population over time?

Populations are composed of genetically varying individuals. Genetic variations are introduced through the random process of mutation. The frequency of genotypes and phenotypes in a population can change over time due to various evolutionary forces. Natural selection increases the frequency of individuals with traits that result in higher reproductive output, and therefore increases the frequency of the alleles associated with those traits. Genetic drift, random fluctuations in allele frequencies, can lead to the loss or fixation of alleles, especially in small populations. Gene flow, the movement of individuals or gametes between populations, can introduce new alleles and alter allele frequencies. Non-random mating can influence the frequency of certain genotypes. These processes collectively shape the genetic makeup of populations and drive evolutionary change.

Sample Learning Objectives:

- Describe the major evolutionary forces (natural selection, genetic drift, gene flow, mutation, and non-random mating) that influence genotype and phenotype frequencies in populations over time.
- Describe the mechanisms by which variation increases and is fixed (or lost) in a population over time.
- Calculate allele frequencies based on phenotypic or genotypic data for a population, and be able to explain the assumptions that make such a calculation possible.
- Model how random mating yields predicted genotype frequencies in Hardy-Weinberg Equilibrium (HWE), and how non-random mating affects allele and genotype frequencies.
- Test whether HWE has been reached in a population.
- Explain why variation, heritability, and differential reproductive success are required for evolution by natural selection
- Explain how natural selection and genetic drift can affect the elimination, maintenance or increase in frequency of various types of alleles (e.g. dominant, recessive, deleterious, beneficial) in a population.
- Interpret experiments to determine the relative influences of genes and the environment on a given phenotype.
- Describe how variation can be measured, and what can be done to distinguish genetic and environmental sources of variation.
- Describe how race is not the same as ancestral lineage/geographic region of origin.
- Analyze how evolutionary theory is informed by genetic evidence, and evaluate the interdisciplinary contributions of molecular genetics, population genetics, and comparative genomics to understanding evolutionary processes such as natural selection, speciation, and adaptation.

Foundational Concept: Methods and Tools in Genetics

Core concept: How do studies in model organisms help us understand fundamental aspects of genetics?

Model organisms provide tractable systems for experimentation that can lead to valuable insights into fundamental genetic principles. These systems create a simplified context for analysis by artificially removing genetic variation in laboratory strains, enabling highly controlled experiments. By studying these organisms, scientists can identify and characterize genes, investigate gene function, and analyze genetic pathways. These findings can create a foundation of knowledge that can be applied toward additional studies in more complex environments and other organisms, including humans. These foundational studies can build an understanding of biological processes with impacts outside the laboratory, such as, development of new treatments for human health conditions, agricultural practices, and environmental conservation, as well as gaining a deeper understanding of the genetic basis of life.

Sample Learning Objectives:

- Justify why information on functions of human genes can often be acquired through studies of model organisms such as yeast, nematode worms, and fruit flies.
- Explain the benefits of using model organisms. Identify specific cases where insights from model organisms have provided crucial insights into the genetic basis of a trait human disease.
- Discuss the limitations of model-organism studies and how naturally-occurring populations differ from laboratory strains.
- Delineate differences among model organisms and describe how the characteristics of individual systems are useful for different types of studies.
- Explain how researchers use model organism databases, such as WormBase and FlyBase, to disseminate information and resources.

Core concept: What experimental approaches are commonly used to analyze gene structure, gene expression, gene function, and genetic variants?

Methods in genetics are constantly changing as new technologies arise. In general, genetic research uses a variety of tools to measure aspects of gene regulation and function and other tools to make changes to genetic information. Broad categories of approaches include determining what DNA sequences are present, making changes to DNA sequences, reducing/eliminating gene function, altering gene expression, examining the molecular characteristics of molecules involved in gene regulation, measuring the abundance of gene products, analyzing the impacts of genetic changes on cell/organism function, determining whether certain features are present across the genome, and studying variation in naturally occurring organisms/populations.

Sample Learning Objectives:

- Understand the principles and applications of CRISPR-Cas9 and other related technologies in genetic research and therapy.
- Describe the forward genetic approach, citing some methods commonly used to create and

screen strains.
- Explain the reverse genetic approach, comparing methods for generating specific mutations in the genome vs. generating phenocopies using techniques such as RNAi or morpholinos.
- Explain the genome-wide approaches to identifying genes, including association studies (GWAS) and knock-out screens.
- Use polygenic risk scores to predict the likelihood of developing complex traits based on multiple genetic variants.
- Navigate and utilize genomic databases such as GenBank, Ensembl, and UCSC Genome Browser to analyze genetic data, including sequence alignment, gene prediction, and phylogenetic analysis.
- Apply the methodologies and applications of Next-Generation Sequencing (NGS) in genomics and personalized medicine.
- Analyze the challenges and advancements in developing ncRNA-based therapeutics, such as siRNA delivery, stability, and reducing off-target effects.

Foundational Concept: Genetics and Ethics

Core concept: How does genetics impact society and society impact genetics?
The field shapes society's views on health, behavior and other social issues. As geneticists and genetics educators, we have a responsibility to acknowledge the past, examine current implications, and look towards the future. The urgency of this responsibility is highlighted by the legacies of eugenics and scientific racism and their present-day manifestations. The roots of genetics are planted in the historical desire of society to predict inheritance of traits. The discipline has been misapplied in unethical attempts to classify and manipulate the inheritance of desirable or undesirable traits to support eugenic ideologies. Scientific findings can be impacted by implicit biases of those involved in the work, including in the selection of research questions and methodologies, interpretation of data, extrapolation of results beyond primary studies, and more. In addition, the way we frame and discuss genetic issues continues to impact societal perception of race and genetic essentialism, sex and gender identities, disability and health impacts, and other topics. Society also influences the work of the field through policies and attitudes related to research funding, access to information, privacy and ownership of genetic information, genetic discrimination, and more. Incorporating training around social and ethical analysis and communication within the genetics community and beyond is an important tool for shaping future attitudes and practices in science and society.

Sample Learning Objectives:
- Explain the legal protections against discrimination based on genetic information in health insurance and employment. For instance, explain how the US law, GINA, prevents employers from using genetic information in hiring decisions and how it may fall short of its intentions.
- Evaluate the ethical implications of using genetic information in forensic science and law enforcement, and propose legal and policy frameworks that balance individual privacy rights with the public interest in criminal justice.
- Discuss differences in ethical considerations in heritable versus somatic gene editing in humans.

As another example, identify ethical considerations that characterize the propagation and use of genetically-modified organisms.

- Apply genetics knowledge to challenge the concept of genetic essentialism in humans, which is a concept that contributes to racist, sexist, and ableist thinking.
- Describe the roles and responsibilities of genetic counselors in healthcare.
- Evaluate the ways that communication about topics in genetics impacts societal attitudes and policies related to those topics.
- Explore the social impact of genetic information, such as the potential for discrimination in insurance, employment, and other areas of life. For example, discuss cases where individuals have faced discrimination based on their genetic predisposition to certain traits.
- Address the legal and ethical issues surrounding the patenting of genetic sequences and technologies. This includes debates over whether genes can be patented and the implications for research and access to genetic tests.
- Discuss the importance of protecting the genetic information of individuals from unauthorized access and ensuring that it is only used for the intended purposes. This includes discussing the risks of data breaches and the importance of secure data storage.
- Discuss how genetics frameworks are often used to describe disability through a medical model, and identify benefits and harms of the medical model relative to other models of disability, such as a social model.